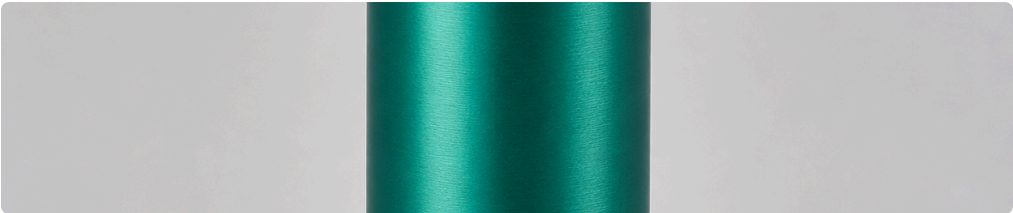
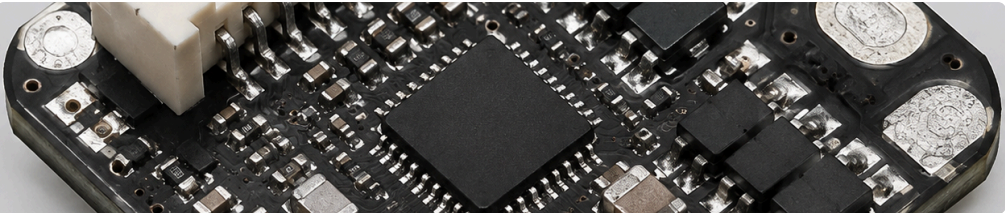


cycler test bench



capacity + cycle count



on-board BMS

DETERMINISTIC MEMORY

state

16 KB

weights

340 KB

state 8 KB × 2 layers

zero malloc

KV-CACHE GROWTH

self-attn

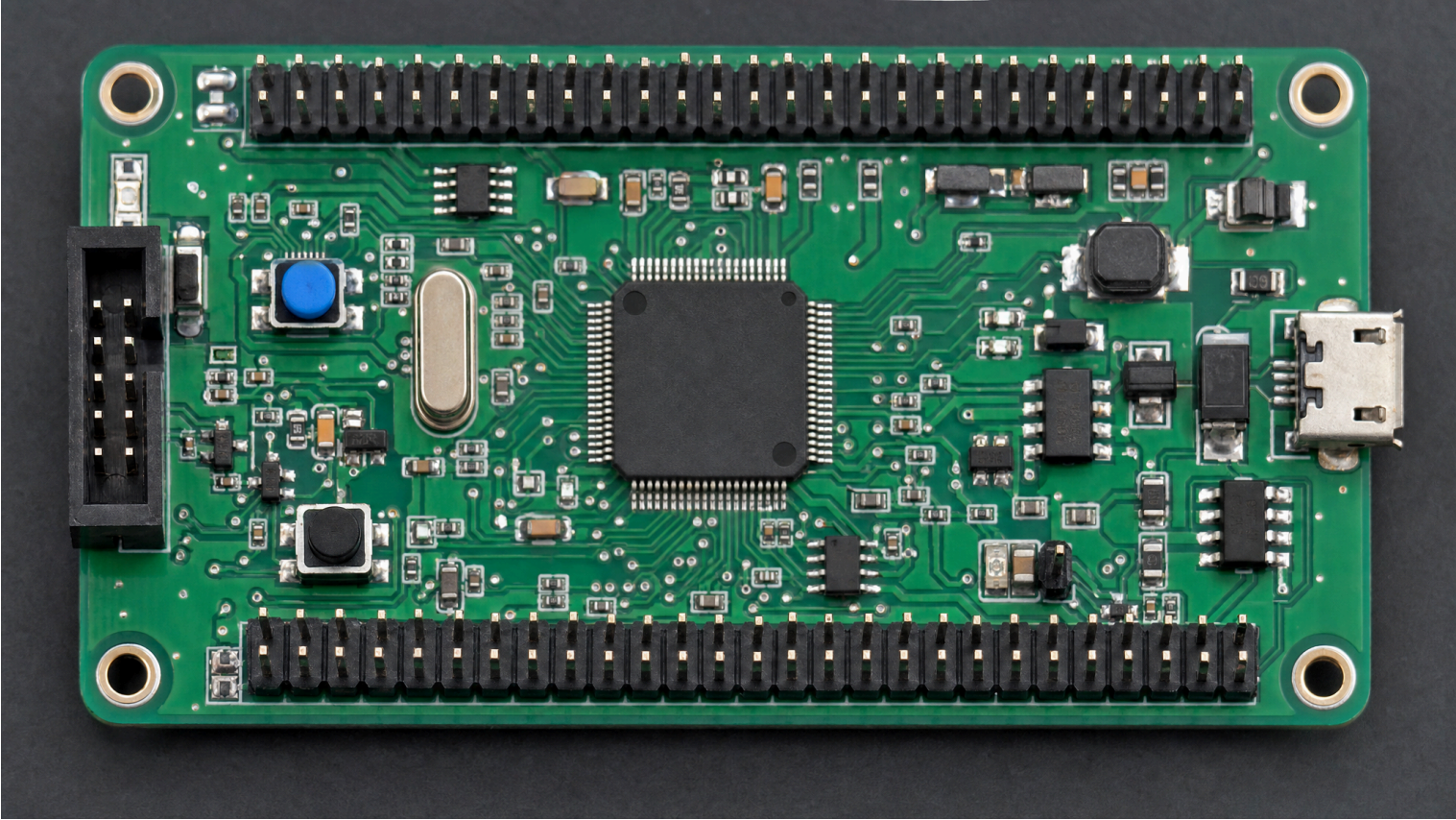
grows O(L)

GDN-2

fixed 16 KB

LIBM · BIT-EXACT ARM vs X86	
expf	identical ✓
logf	identical ✓
sqrtf	identical ✓
erff	identical ✓

ON-BOARD MCU · FULL NETWORK IN C



INT8 QUANTIZATION

fp32 → INT8

−75%

accuracy

AE 2.2 → 2.3

P2.5

P50

P97.5

OUTPUTS

calibrated interval → RUL

capacity-only input

REAL LATENCY (C · QEMU-MEASURED)

Cortex-M3

428 ms @ 25 MHz

STM32F4

8–12 ms @ 168 MHz

< 3 × 10^{−6}

full-scan diff vs PyTorch

no heap · no float drift · static allocation